

Question	Marking Guidance	Mark	Comments
02.1	Box 2. An inversion will result in a change in the number of DNA bases.	1	Reject if more than one box with tick. Ignore crossed-out ticks
02.2	1.(Increased) methylation (of tumour suppressor genes); 2.Mutation (in tumour suppressor genes); 3. Tumour suppressor genes are not transcribed/expressed OR Amino acid sequence/primary structure altered; 4. (Results in) rapid/uncontrollable cell division;	3 max	1. Accept abnormal methylation or hypermethylation 1. Ignore decreased acetylation of histones 3. Accept mRNA for transcription/transcribed 3. Accept tertiary structure altered 3 Accept different amino acid 3. Ignore reference to protein not being formed 4. Accept cell division cannot be regulated 4 Ignore growth
02.3	1. Correct answer of $1.9/1.93 \times 10^{25} = \mathbf{2 \text{ marks}};$ 2. Incorrect answer but shows $84 = \mathbf{1 \text{ mark}}$ OR $28 \times 3 = \mathbf{1 \text{ mark}}$ OR Incorrect answer but shows 672 divided by 8 = 1 mark;	2	1. Accept $2 \times 10^{25} = \mathbf{2 \text{ marks}}$ 1. Ignore any numbers after 1.93

